sumers a variety to choose from.

In fact, VIA was the first to realise the performance gain of DDR RAM and brought it to the Intel platform before Intel could do so with their chipsets.

Intel, with their 810 and 815 chipsets for the Pentium 3 processor, made on-board graphics popular, especially in India where the majority of the PCs sold have on-board graphics. The 845 chipset further improved graphics performance. VIA and SiS did bring good competing products, but lost ground to Chipzilla's marketing muscle.

Greater demand for on-board graphics spurred Intel to introduce the i865 and i875 chipsets, the former targeted towards price-sensitive users, the latter meant as the performance chipset.

By this time, graphics card leaders nVidia and ATi realised the potential market Intel had tapped into by including a graphics chip on board their chipsets. This heralded the birth of the ATi RS350 PRO chipset with better onboard graphics than Intel's chipset.

Last year, Intel introduced the i915 and 925X chipsets, which were completely new platforms designed from the ground up, offering new features and revising some specifications to handle today's media-savvy world.

What's On Offer

If you want to buy an Intel-based system, the market offers motherboards based on the i845, i865, i875, i915, i925 and RS350 chipsets. Asus, MSI, Gigabyte, Mercury, HIS, Krypton are the brands you will come across. Let's understand the pros and cons of each chipset.

i845

The 845 chipset is the oldest available chipset in the market. It is generally used in motherboards that do not cost more than Rs 4,000, and hence, we will treat it as out of the scope of this article, since we are primarily looking at mid-range motherboards.

Systems based on this chipset are good for Internet browsing and Word processing—nothing too computationally intensive.

i865 and i875

Both these chipsets are relatively new and have enough juice to hold their own against newer Intel chipsets. However, since they lack PCI-Express, they may become obsolete too soon. The difference between the two chipsets is that the i865 is aimed at the value-conscious buyer looking for an onboard graphics solution, whereas the i875P chipset offers better performance.

One point to remember is that motherboards based on these chipsets use the older Socket 478 processors. In fact, ATi's RS350 is also built around the Socket 478 processor, and hence the new LGA processors won't fit on these motherboards.

i915 and i925

This brand-new chipset family from Intel is loaded with features—some new, some logical and some revised to meet current standards. But the highlight is the introduction of the PCI-Express bus for discrete graphics cards and other peripherals. Nearly all motherboards based on these two chipsets will have an x16 PCIe slot for a graphics card and two or more x1 PCIe lanes in addition.

These chipsets also support the new DDR II standard, the memory runs at 533MHz—133 MHz faster than the fastest DDR memory. On the integrated graphics front, the 915 chipset boasts of DirectX 9.0 compliance, and is capable of hooking up to an HDTV or to widescreen LCD monitors.

Most importantly, improved onboard graphics have been introduced so that a platform based on this chipset complies with Microsoft's upcoming Longhorn OS' minimum requirements.

On the storage side, the new chipsets have legacy support for four SATA drives and one ATA drive. If the motherboard uses the ICH6R Southbridge, it supports Intel's Matrix RAID technology too. Eight USB2.0 ports and a dedicated Gigabit network bus is an integral part of all motherboards, unless the motherboard manufacturer decides to cut down on cost.

The introduction of High-Definition Audio (Azalia) will be music to your ears! The new standard improves upon the older AC'97 and takes it further: you can connect 7.1 speakers directly to your motherboard; there's no need for an additional 7.1-channel soundcard. Most motherboards also have 'jack sensing' enabled, which ensures that